Quantification: 542_z18
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	10.3636	10	5.0847	count
ring_length_r30_px	headline	452.727	492	162.089	px
ring_length_r60_px	reported	1131.82	1196	370.879	px
owned_length_px	ownership-dependent	2686.38	2973.57	986.691	px
edge_count	ownership-dependent	118.273	131	46.7185	count
mean_edge_length_px	ownership-dependent	23.8444	22.8151	4.2105	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	11901.2	12118	3231.07	px^2
ring_area_r60_px2	normalised	28359.4	29514	6840.82	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0370072	0.0382605	0.00697851	px^-1
ring_density_r60	normalised	0.0390216	0.0381881	0.00749029	px^-1
roots_per_100px_perimeter	normalised	2.9114	2.7581	0.8862	per 100 px
ring_attached_length_r30_px	network v2	388.636	397	163.208	px
ring_attached_length_r60_px	network v2	924.455	961	379.219	px
ring_length_w_r30_px	network v2	507.273	547.475	185.786	px
ring_length_w_r60_px	network v2	1278.7	1344.87	424.905	px
sholl_crossings_r10	network v2	11.1818	12	4.8748	count
sholl_crossings_r20	network v2	15.0909	15	6.0408	count
sholl_crossings_r30	network v2	16.1818	17	5.3445	count
ring_width_mean_r30_px	width	7.5225	7.4427	0.593	px

Per field
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measure	value	units
analysed_area_px2	1.00717e+06	px^2
total_skeleton_length_px	38252	px
canalicular_length_density_per_px	0.0379798	px^-1
junction_count	828	count
junction_density_per_px2	0.00082211	px^-2
lacuna_count	13	count
lacunae_per_px2	1.291e-05	px^-2
skeleton_component_count	241	count

mean_component_length_px	158.722	px
median_component_length_px	24	px
field_length_density_w_per_px	0.042932	px^-1
field_density_without_flagged_per_px	0.0379437	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	372	px
unattached_component_count	22	count
canalicular_length_density_attached_per_px	0.0379798	px^-1
field_length_density_with_unattached_per_px	0.0383491	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2207	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	36730	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False
13								1015.14	933.902	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	226	591	1137.85	65	17.5054	7.3669
2	20	687	1528	2986.4	151	19.7775	7.3988
3	8	471	1199	3353.82	147	22.8151	7.9167
4	17	541	1312	3609.21	160	22.5576	7.3242
5	11	505	1180	2370.6	127	18.6661	8.3498
6	10	440	1196	3371.92	143	23.5798	7.5592
7	13	605	1554	2589.92	128	20.2338	8.382
8	8	371	999	2147.4	86	24.9698	7.7776
9	12	512	1161	2973.57	131	22.699	6.8416
10	8	492	1367	4072.25	161	25.2935	7.4427
11	4	252	658	1176.58	35	33.6165	7.3845

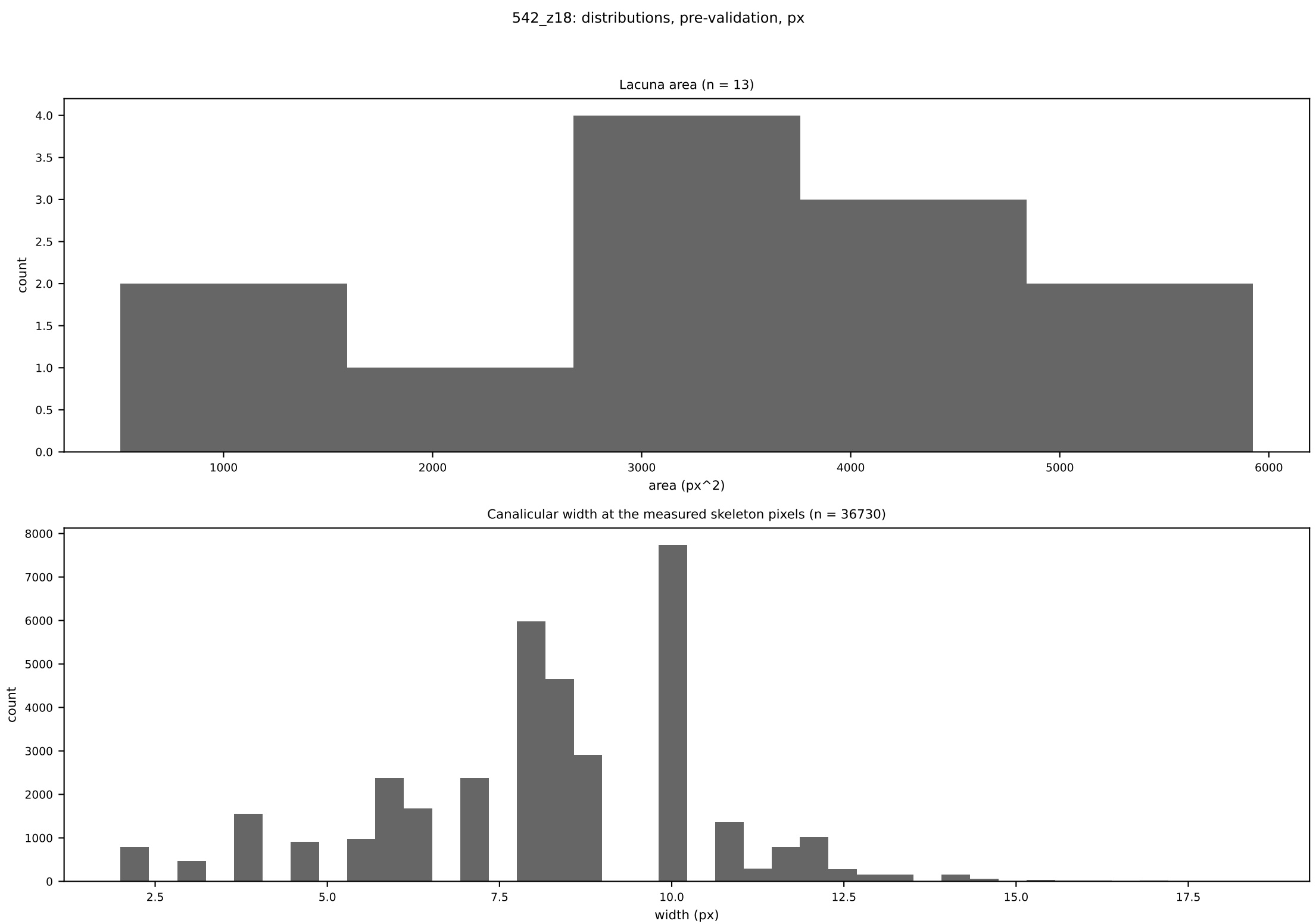
12	3	104	296	898.536	32	28.0793	6.3702	
13	0	0	9	0	0	0		

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	142.811	4575	11641	0.673	0.6207	0.0493989	0.0507688	2.1007
2	625.725	16860	35430	0.9542	0.8861	0.0407473	0.0431273	3.1963
3	377.019	13152	31926	1	1	0.035812	0.0375556	2.1219
4	337.706	12118	29514	1	1	0.0446443	0.0444535	5.034
5	371.806	13199	31872	1	0.9956	0.0382605	0.0370231	2.9585
6	308.593	11379	28385	1	1	0.0386677	0.0421349	3.2405
7	382.877	12760	30245	1	1	0.0474138	0.0513804	3.3953
8	306.492	10697	26160	1	1	0.0346826	0.0381881	2.6102
9	437.262	14880	35351	1	1	0.0344086	0.0328421	2.7444
10	326.836	11837	29271	1	1	0.0415646	0.0467015	2.4477
11	263.48	9968	22850	0.9798	0.8783	0.0252809	0.0287965	1.5181
12	108.77	4063	10949	0.8837	0.75	0.0255969	0.0270344	2.7581
13	59.8995	971	1939	0.6158	0.5712	0	0.00464157	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	219	463	256.995	665.316	4	7	6
2	667	1237	777.097	1741.12	22	24	23
3	397	925	527.546	1344.87	13	15	19
4	450	971	600.203	1499.15	12	18	19
5	351	955	563.889	1334.19	7	16	17
6	366	961	495.404	1371.31	10	14	16
7	581	1493	684.357	1745.91	13	24	20
8	318	862	408.25	1114.22	10	13	15
9	462	961	587.801	1332.69	14	13	16
10	420	1280	547.475	1528.64	12	17	20
11	167	303	278.865	733.47	7	9	8
12	96	221	109.113	320.066	3	3	5
13	0	0	0	10.0711	0	0	0


542_z18: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36730)
count
width (px)